

GALATI, Romania

WMO: 153100

Lat: 45.4731N

Lon: 28.0322E

Elev: 231

StdP: 14.57

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	13.8	3.6	6.7	11.1	8.2	8.5	16.7	22.9	25.5	20.2	27.9	7.2	20	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	93.1	70.6	90.2	69.5	87.3	68.5	74.0	85.6	72.3	84.0	70.8	82.4	5.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	113.3	79.3	68.7	106.2	77.3	67.1	100.4	75.9	37.7	85.7	36.1	83.8	34.8	82.3	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.3	14.4	2.9	97.7	5.4	3.1	-1.0	99.9	-4.2	101.8	-7.3	103.5	-11.2	105.8		
(5)			2.2	77.2	5.3	3.5	-1.6	79.7	-4.7	81.8	-7.7	83.8	-11.6	86.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	29.9	34.3	43.1	54.0	64.4	71.3	75.3	74.7	65.1	53.9	43.9	33.3		
	DBStd	17.61	8.92	9.72	8.56	7.00	6.33	5.70	5.11	5.82	6.65	7.82	8.50	8.81		
	HDD50	2143	624	444	244	39	1	0	0	0	0	51	220	520		
	HDD65	5119	1089	861	680	335	88	12	1	4	79	352	634	984		
	CDD50	3493	0	3	30	160	448	640	783	765	454	171	37	2		
	CDD65	993	0	0	0	6	69	202	320	305	83	8	0	0		
	CDH74	8910	0	0	2	58	577	1729	3014	2874	612	44	0	0		
	CDH80	3353	0	0	0	11	127	596	1236	1217	163	3	0	0		
	WSAvg	6.3	6.9	7.2	7.6	7.0	6.2	5.9	5.7	5.4	5.6	5.7	6.1	6.7		
	PrecAvg	19.1	1.1	1.1	1.1	1.5	1.9	2.6	1.9	1.7	1.8	1.3	1.4	1.3		
(15) Precipitation	PrecMax	32.8	5.7	3.7	3.4	3.3	5.3	6.1	8.2	5.7	5.4	5.4	4.4	5.4		
	PrecMin	14.3	0.0	0.2	0.0	0.0	0.2	0.2	0.0	0.2	0.0	0.1	0.2	0.0		
	PrecStd	4.1	1.0	0.8	0.8	0.7	1.1	1.3	1.7	1.4	1.5	1.3	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	61.6	70.3	81.0	87.7	93.3	97.5	97.5	89.7	79.2	69.2	56.5		
		MCWB	46.3	51.2	53.6	61.9	65.3	71.1	69.7	71.4	66.6	64.0	59.1	50.9		
	2%	DB	47.4	55.3	65.3	74.1	83.5	89.5	93.1	93.3	84.5	73.9	63.1	52.0		
		MCWB	43.1	47.2	51.8	58.3	64.8	69.4	70.6	70.9	65.2	61.5	55.6	48.2		
	5%	DB	43.9	50.7	60.6	69.9	80.2	86.4	89.9	90.3	80.7	69.6	58.8	47.8		
		MCWB	41.0	45.1	49.5	55.9	63.3	68.3	70.1	69.4	64.2	59.6	53.7	44.6		
	10%	DB	40.9	46.3	56.0	66.1	76.9	83.3	86.8	87.0	77.1	65.4	55.5	44.0		
		MCWB	38.9	42.3	46.8	54.3	62.0	67.1	69.1	68.4	63.0	57.4	51.9	41.6		
	0.4%	WB	47.1	52.1	55.4	62.9	69.3	74.6	76.0	78.0	70.8	67.0	59.9	52.0		
		MCDB	51.1	59.3	67.9	77.2	81.0	86.4	88.5	90.4	80.3	75.5	66.8	55.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.8	48.3	52.5	60.0	67.0	72.1	74.0	74.3	68.4	63.4	56.9	48.5		
		MCDB	47.0	54.4	63.3	71.7	79.1	83.8	85.6	86.0	76.9	70.2	62.0	52.2		
	5%	WB	41.3	45.5	50.2	57.5	65.2	70.5	72.5	72.0	66.5	60.7	54.5	45.0		
		MCDB	43.4	49.8	59.2	67.0	76.5	82.5	84.2	83.7	75.6	67.7	58.1	47.5		
	10%	WB	38.9	42.8	47.8	55.4	63.4	68.9	71.0	70.2	64.7	58.4	52.2	41.9		
		MCDB	40.8	46.3	54.4	63.7	73.6	80.1	82.6	81.8	73.9	64.4	55.4	43.9		
	5% DB	MDBR	9.7	12.1	15.6	17.6	19.1	19.0	19.8	20.1	18.3	15.3	11.5	9.6		
		MCDBR	13.0	19.0	23.5	23.5	23.6	22.9	23.6	24.3	23.2	20.7	16.5	13.5		
		MCWBR	9.6	12.1	12.3	10.9	9.9	8.7	8.0	8.4	8.6	10.3	9.8	10.0		
		MCWBR	12.1	17.3	21.6	21.0	21.5	20.9	20.7	21.3	20.1	18.8	14.4	13.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.4	12.0	12.2	10.5	9.9	9.1	8.4	8.8	9.1	10.2	8.9	10.1		
	Clear-Sky Solar Irradiance	taub	0.339	0.370	0.404	0.456	0.429	0.428	0.446	0.460	0.420	0.387	0.363	0.339		
		taud	2.289	2.224	2.183	2.056	2.219	2.256	2.213	2.164	2.237	2.311	2.324	2.309		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	236	251	259	255	267	267	260	251	251	242	226	222		
		Edh at Noon	26	34	40	50	44	42	43	44	38	30	25	23		
(44) All-Sky Solar Radiation	RadStd		395	652	1038	1468	1853	1964	1978	1782	1293	826	442	333		
			58	84	101	141	134	130	127	106	111	82	54	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.17	N/A	N/A	+1.26	N/A	N/A	-221	-317	+236	+162
(47) Regional (47 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	N/A	-232	+166	+104

Nomenclature: See separate page